FBK-HLT-time: a complete Italian Temporal Processing system for EVENTI-EVALITA 2014

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End-to-end system

- End-to-end system for temporal processing of Italian texts.
- Subsystems first developed for English as part of the NewsReader project and adapted to Italian.
- Follow It-TimeML specifications.

Tools

TextPro	a suite of NLP tools for processing English and Italian texts (http://textpro.fbk.eu/)
Yamcha	a text chunker which uses Support Vector Machines algorithm (http://chasen.org/ taku/software/yamcha)

Examples

1. **Time Expression (timex)** recognition and normalization
2. **Event** extraction and classification
3. **Temporal Relation** extraction and classification

Time Expression Recognition

1. Classification task using the IOB2 tagging format
2. Rules for the identification of non text-consuming TIMEX3

Features:

- Token's text, lemma, part-of-speech, chunk, entity's type
- Whether the token match regular expression patterns: units, part of a day, time, etc.
- All of the above features for the preceding two and following two tokens
- The preceding two labels tagged by the classifier

Event Extraction

Classification task in two steps: identification of event extents and classification of the identified events.

Features:

- Token's lemma, stem (using Snowball Italian stemmer), part-of-speech, chunk, simplified part-of-speech, verb's tense
- Whether the token is part of a time expression, entity's type
- Token's WordNet domain (using MultiWordNet)
- Token's suffix: -zione, -mento, -tura and -aggio
- The frequency of the token's appearance in an event extent
- Token's derivative (using the derIvaTario resource)
- Some of the above features for the two preceding tokens and the following one
- The preceding 3 labels tagged by the classifier

Temporal Relation Extraction

Temporal Link Identification:

1. all combination of event/event and event/timex pairs within a sentence are considered
2. classification of pairs in two classes: REL or O

Temporal Relation Classification: classification of the pairs detected in the previous step.

- In building the classifier for event/event pairs, only the frequent relation types are considered: BEFORE, AFTER, INCLUDES, etc.
- Addition of inverse relations (e.g. BEFORE/AFTER) to extend the size of the corpus
- Some patterns explained in the guidelines are used to classify event/timex pairs (EVENT + dal + DATE$_{t}$*ype* →relType=BEGUN_BY)

Features:

- String and grammatical features: tokens, lemmas, PoS, etc.
- Textual context: pair order, entity distance, etc.
- Entity attributes: event attributes (class, tense, aspect and polarity) and timex type.
- Dependency information: dependency relation type between the two entities, etc.
- Temporal signals: tokens of temporal signals occurring around the two entities, etc.

[Paramita Mirza and Sara Tonelli. 2014. Classifying Temporal Relations with Simple Features. In Proceedings of EACL2014.]

Time Expression Normalization

TimeNorm: time expression normalizer for English developed by Bethard (2013), based on synchronous context free grammars. It parses time expressions, and given an anchor time return all possible normalizations following TimeML specifications.

Adaptation to Italian:

- translation and modification of the existing English grammar
- modification of TimeNorm code in order to support Italian language specificity (normalization of accented letters, unification of articles and articulated prepositions, etc.)

Pre-processing: processing of time expressions composed by only one or two digits, and addition of either a unit or a name of month (inferred from a nearby timex or from the document creation time).

Post-processing: selection of one of the value returned by the system using simple rules.

System Performance

Task A: Time expression recognition and normalization

	F1	R	P	Strict F1	Strict R	Strict P	type F1	value F1
Main task	**0.886**	0.841	0.936	**0.827**	0.785	0.873	0.800	0.665
Pilot task[1]	**0.870**	0.794	0.963	**0.746**	0.680	0.825	0.678	0.475

Task B: Event extraction

	F1	R	P	Strict F1	Strict R	Strict P	class F1
Main task	**0.884**	0.868	0.902	**0.867**	0.850	0.884	0.671
Pilot task	**0.843**	0.793	0.900	**0.834**	0.784	0.890	0.604

Task D: Temporal Relation classification

		F1	R	P	Strict F1	Strict R	Strict P
Main task	Run1	**0.736**	0.731	0.740	**0.731**	0.727	0.735
	Run2*	**0.738**	0.733	0.742	**0.733**	0.729	0.737
Pilot task	Run1 & 2	**0.588**	0.588	0.588	**0.570**	0.570	0.570

Task C: Temporal Relation identification and classification

	F1	R	P	Strict F1	Strict R	Strict P
Main task	**0.264**	0.238	0.296	**0.341**	0.308	0.381
Pilot task	**0.185**	0.139	0.277	**0.232**	0.173	0.349

[1] Pilot task: temporal processing of historical texts

Conclusion

- Without any specific adaptation to historical text, the system yields comparable results on the main task and pilot task.
- The system will be included in the TextPro tools suite (http://textpro.fbk.eu/), both for Italian and English.